

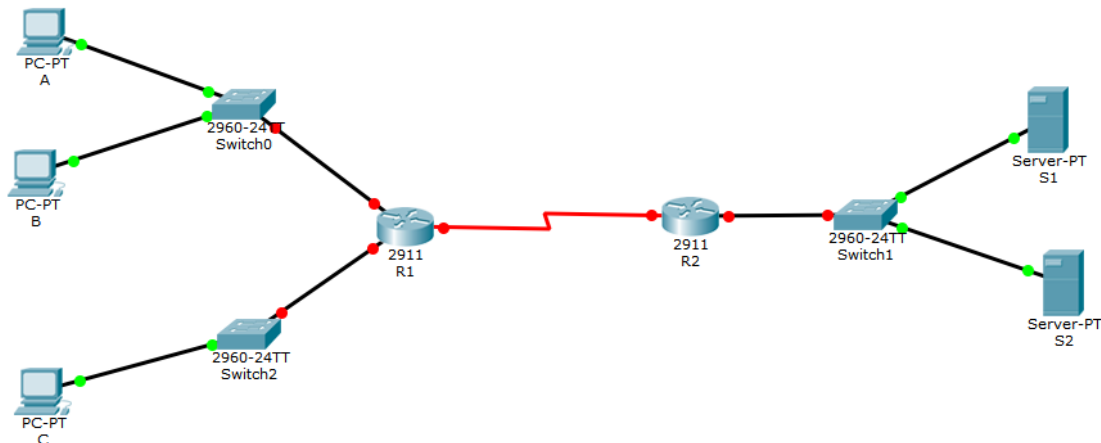


Lab 6 – OSPF

The goal of Lab is to achieve a dynamic routing using OSPF. Two topologies will be studied.

Part1: small network

Consider the following topology



1st step: Create the topology.

Use 2 routers of type 2911.

2nd step: Configuration of IP addresses and Gateways.

- Host A : 10.1.1.1/24. Gateway : 10.1.1.3
- Host B : 10.1.1.2/24. Gateway : 10.1.1.3
- Interface G0/0 of R1 : 10.1.1.3/24
- Host C : 10.3.3.3/25. Gateway : 10.3.3.1
- Interface G0/1 of R1 : 10.3.3.1/25
- S1 : 10.2.2.1/24. Gateway : 10.2.2.3
- S2 : 10.2.2.2/24. Gateway : 10.2.2.3
- Interface G0/0 of R2 : 10.2.2.3/24
- Interface S0/3/0 of R1: 10.4.4.1/30
- Interface S0/3/0 of R2 : 10.4.4.2/30

3rd step: Test the connectivity between the machines of the same subnet by using “ping”.

- From Host A to Host B.
- From Host C to the interface G0/1 of R1.
- From S1 to S2.
- From the interface S0/3/0 of R1 to the interface S0/3/0 of R2.

Note that ping between hosts of different subnets cannot succeed because routers do not have entries in their routing table.

4th step: Activation of OSPF on the interfaces of the two routers R1 and R2 (area 0).

Use the following configuration:

R1:

```
Router#configure terminal
Router (config)#router ospf 1
Router (config-router)#network 10.0.0.0 0.255.255.255 area 0
Router (config-router)#end
```

R2:

```
Router#configure terminal
Router (config)#router ospf 1
Router (config-router)#network 10.0.0.0 0.255.255.255 area 0
Router (config-router)#end
```

Make sure that the neighboring relationships have been created between R1 and R2.

R1:

Router#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.4.4.2	0	FULL/ -	00:00:39	10.4.4.2	Serial0/3/0

R2:

Router#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.4.4.1	0	FULL/ -	00:00:34	10.4.4.1	Serial0/3/0

6th step: Connectivity test

Perform connectivity tests and make sure that you can ping between hosts of different subnets.

- From Host A to Host C.
- From Host C to S1.
- From Host A to S1.

Part2: Extended network

Consider the following topology

Repeat the 6 steps to configure OSPF on the following topology.

